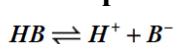


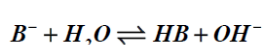
$$\begin{aligned} \text{pH} &= -\log[\text{H}^+] & \text{pH} + \text{pOH} &= \text{p}K_w = 14 = \text{p}K_a + \text{p}K_b \\ \text{pOH} &= -\log[\text{OH}^-] \\ \text{p}K_a &= -\log(K_a) & \text{p}K_b &= -\log(K_b) \end{aligned}$$

- **Strong acids:** $\text{HA}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{A}^-(\text{aq})$
 - 1) Strong acids completely ionized to ions in aqueous solution;
 - 2) Common strong acids: HCl , HBr , HI , HNO_3 , HClO_3 , HClO_4 , H_2SO_4 ;
 - 3) For the monoprotic strong acids: $[\text{H}^+] = [\text{acid}]$;
- **Strong Bases:** $\text{M}(\text{OH})_n(\text{aq}) \rightarrow \text{M}^{n+}(\text{aq}) + n\text{OH}^-(\text{aq})$
 - 1) Strong bases completely ionized to ions in aqueous solution;
 - 2) Common strong bases are the ionic hydroxides of the alkali metals (NaOH , KOH) and the ionic hydroxides heavier alkaline earth metals ($\text{Sr}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$);
 - 3) Strongly basic solutions are created by certain substances (Na_2O , CaO) that react with water to form OH^- (aq);
- **Weak acids:** $\text{HA}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{A}^-(\text{aq})$
 - 1) Weak acids only partially ionized to ions in aqueous solution;
 - 2) Acid-dissociation constant: $K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$; 酸常数
 - 3) The greater the value of K_a , the stronger is the acid;
- **Weak bases:** $\text{B}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HB}^+(\text{aq}) + \text{OH}^-(\text{aq})$
 - 1) Weak bases only partially ionized to ions in aqueous solution;
 - 2) Base-dissociation constant: $K_b = \frac{[\text{HB}^+][\text{OH}^-]}{[\text{B}]}$; 碱常数
 - 3) The greater the value of K_b , the stronger is the base;
 - 4) Common weak bases are neutral substances that have an atom with a nonbonding pair of electrons that can accept a proton (NH_3) and anions of weak acids (ClO^-);

- **Relationship between K_a and K_b .** (for conjugate acid-base pair)



$$K_a = \frac{[\text{H}^+][\text{B}^-]}{[\text{HB}]}$$



$$K_b = \frac{[\text{HB}][\text{OH}^-]}{[\text{B}^-]}$$

$$K_a K_b = \frac{[\text{H}^+][\text{B}^-]}{[\text{HB}]} \times \frac{[\text{HB}][\text{OH}^-]}{[\text{B}^-]} = [\text{H}^+][\text{OH}^-] = K_w$$

$\times$ (อัตรา) $> 5\%$ ของ สารละลาย $\rightarrow$ ใช้ quadratic

$M = \text{original acid conc}$, $a = \text{conc of } \text{H}_3\text{O}^+ \text{ at equilibrium}$

$\frac{a^2}{M-a} = K_a \rightarrow \frac{a^2}{M} = \frac{K_a}{a} \rightarrow \frac{M}{a} = \frac{M-a}{a} + 1 = \frac{a+K_a}{K_a}$

$\frac{a}{M} = \frac{K_a}{a+K_a}$

- **Percent Ionization:** $\text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH}^-$ endothermic $T \uparrow \rightarrow K_w \uparrow$ (reaction shifts right) $\rightarrow \text{pH} \downarrow$

$$\text{Percent ionization} = \frac{\text{concentration of ionized HA}}{\text{original concentration of HA}} \times 100\% = \frac{[\text{H}^+]_{\text{equilibrium}}}{[\text{HA}]_{\text{initial}}} \times 100\%$$

电离度

- 1)
- 2) Indicate the strength of a weak acid, the stronger the acid, the greater the percent ionization. $K_a \uparrow$

- **Polyprotic Acids:**

- 1) Acids that have more than one ionized H atom are polyprotic acids;
- 2) $\text{H}_2\text{A}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HA}^-(\text{aq})$ K_{a1}
 $\text{HA}^-(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{A}^{2-}(\text{aq})$ K_{a2} $\text{pH} \uparrow \text{ then } K_{a1}$
- 3) K_{a2} refers to the equilibrium involving removal of the second proton of a polyprotic acid; $K_{a2} \ll K_{a1}$
- 4) It is always easier to remove the first proton from a polyprotic acid than to remove the second;

- **Acid-Base Properties of Salts:** we assume that any salt dissolved in water is completely dissociated, so the acid-base properties of salt solutions are due to the behavior of the cations and anions.

- **Anions:**

- 1) Anions of strong acids are neutral (Cl^-);
- 2) Anions of weak acids are conjugate bases and create OH^- in water (CH_3COO^-);

(ข้อสอบเก่า ๆ ที่เคยทำ)

แก๊ส + กรดแอค + เบสแอค $\Rightarrow$ เกลือ
 กรดแอค + เบสแอค $\Rightarrow$ เกลือ
 กรดแอค + เบสแอค $\Rightarrow$ เกลือ
 กรดแอค + เบสแอค $\Rightarrow$ เกลือ K_a, K_b

- 3) Protonated anions from polyprotic acids can be acids or bases (if $K_a > K_b$, the anion will be acidic; if $K_b > K_a$, the anion will be basic);

● **Cations:**

- 1) Group I or Group II metal cations are neutral (Ca^{2+});
- 2) Polyatomic cations contain one or more protons can be considered the conjugate acids of a weak base (NH_4^+);
- 3) Transition and post-transition metal cations are acidic;

Cation	K_a
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}

Charge ↑ Acidity ↑

● **Factors that Affect Acid Strength:**

- 1) **Polarity of H-A bond:** A molecule containing H will act an acid only if the H-A bond is polarized such that the H atom has a partial positive charge; $\text{H} \rightarrow \text{A}$
- 2) **Strength of H-A bond:** Weaker H-A bonds can be broken more easily, making the acid stronger;
- 3) **Stability of conjugate base:** In general, the greater the stability of the conjugate base, the stronger the acid;

The strength of an acid is often a combination of **all three factors**.

● **Binary Acids:** consist of H and one other element.

- 1) Within a group, the **strength of H-A bond** is the major factor to determine acid strength, acidity increase as the bond strength decreases;
- 2) Within a period, **bond polarity** is the major factor to determine acid strength, acidity increases as the electronegativity of A increases;

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● **Oxyacids:** Y-OH, consist of H, O, and one other element.

$\text{YOH} \rightarrow \text{YO}^-$ for nonmetals Y

- 1) If Y is a metal, the compounds would behave as a base (NaOH);
- 2) If Y is a nonmetal, the compounds would be acidic or neutral. Dissociation of an oxoacid involves breaking an O-H bond. Any factor that weakens the O-H bond or increases its polarity increases the strength of the acid.
 - a) For oxoacids that contain the same number of OH groups and the same number of O atoms, acid strength increases with increasing electronegativity of Y. e.g.,
 - ✓ In a same group, the acidity of acid follows the order: $\text{HClO} > \text{HBrO} > \text{HIO}$;
 - ✓ In a same period, the acidity of acid follows the order: $\text{HClO}_4 > \text{H}_2\text{SO}_4 > \text{H}_3\text{PO}_4$;
 - b) For oxoacids that contain the same atom Y but different amounts of oxygen atoms, acid strength increases with increasing oxidation number of Y. ($\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$);

Y's EN ↑ acidity ↑
 $\rightarrow \text{YO}^-$ more stable

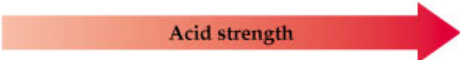
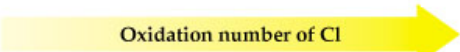
ขาล้าง

O มาก $\Rightarrow$ เป็นกรดแอค

#O ↑ $\rightarrow$ กระจาย $e^- \rightarrow \text{H}^+$ แล้วยึดกัน
 Acid strength

	H-O-I	<	H-O-Br	<	H-O-Cl
Dissociation constant, K_a	2.3×10^{-11}		2.0×10^{-9}		3.5×10^{-8}
Electronegativity	2.5		2.8		3.0

Electronegativity of Y

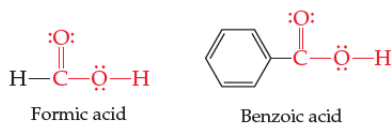
							
	$\text{H}-\text{O}-\text{Cl}$	$<$	$\text{H}-\text{O}-\text{Cl}-\text{O}$	$<$	$\text{H}-\text{O}-\overset{\text{O}}{\underset{\text{O}}{\text{Cl}}}-\text{O}$	$<$	$\text{H}-\text{O}-\overset{\text{O}}{\underset{\text{O}}{\text{Cl}}}-\text{O}$
Name of acid	Hypochlorous		Chlorous		Chloric		Perchloric
Dissociation constant, K_a	3.5×10^{-8}		1.2×10^{-2}		~ 1		Very large
Oxidation number of Cl	+1		+3		+5		+7
							

羧酸

- **Carboxylic Acids:** organic acids that contain the $-\text{COOH}$ group;

Factors contributing to their acidic behavior:

- 1) The additional O atom attached to the C of the carboxyl group draws electron density from the O-H bond, increasing the polarity and helping to stabilize the conjugate base;



- 2) The conjugate base of a carboxyl acid can exhibit resonance, which contributes to the stability of the anion by spreading the negative charge over several atoms;

